

Xiyan Zhou

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EDUCATION

University of Michigan, Ann Arbor

Ann Arbor, MI

B.S. Computer Science & B.S. Mathematics (Double Major)

GPA: 3.8 / 4.0

Coursework: Measure Theory, Convex Optimization, Bayesian Machine Learning, Numerical Linear Algebra, Real Analysis, Probability Theory, Stochastic Processes, Machine Learning, Operating Systems, Data Structures & Algorithms, Combinatorics, Discrete Mathematics

SKILLS

Programming Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript

Frameworks & Tools: JAX, Flax, PyTorch, NumPy, scikit-learn, L^AT_EX, Git, Bash, Linux

ML & Math: Reinforcement Learning, Bayesian Methods, Convex Optimization, CNNs, Statistical Modeling

Languages: Mandarin (native), English (proficient)

RESEARCH

Motion Planning via Contrastive Reinforcement Learning

Research Assistant — advised by Kellen Kanarios (Lei Ying Lab)

Jan 2026 – Present

- Studying goal-conditioned contrastive RL (CRL) with a Gumbel-Softmax actor for discrete action spaces, extending JaxGCRL to categorical logits; diagnosed and fixed a critical action-encoding bug, raising eval success from 3% to 85% (peak 91%) on gridworld tasks.
- Extended CRL to abstract, position-invariant inventory-collection goals via thermometer encoding, reaching 68% success with graceful difficulty degradation and zero-shot generalization to out-of-distribution moving-goal targets.
- Designed a controlled ablation testing a claim from a recent ICML 2026 paper (LEO) that CRL “cannot natively handle” procedurally generated, abstract-goal settings; reproduced the predicted failure mode (success collapsing 68%→8% under layout randomization).
- Implementing double Q-learning, with a hierarchical extension planned; benchmarking agents across Navix and custom grid-world environments, targeting a workshop submission (NeurIPS/AAAI workshops).

AI-Driven Human Digital Twin Study (IOE SURE Program)

Undergraduate Research Assistant — advised by Prof. Jingwen Hu

May 2026 – Present

- Developing the lower extremity muscle segmentation component of the HERMES human body model; identified that TotalSegmentator, the standard public CT segmentation tool, only segments thigh muscles into broad compartments rather than individual muscles.
- Determined that compartment-level segmentation is insufficient for this application, since individual muscles within a compartment differ in fiber direction (relevant to force analysis) and cannot be separately evaluated for cross-sectional area.
- Hand-labeling individual thigh muscles from CT data to build a training set, and preparing to fine-tune the nnU-Net architecture underlying TotalSegmentator for automatic individual-muscle segmentation, potentially aided by transfer learning.

PROJECTS

Replicating and Extending SimGRL: Robustness in Visual RL under Dynamic Scene Perturbations

- Replicated the SimGRL visual RL framework (PyTorch, DMControl) and designed four data augmentation extensions (Irregular-SRO, Multi-Source SRO, Hybrid-SRO, Mask-Guided); ablations on the Great Lakes HPC cluster showed Hybrid-SRO significantly reduced performance variance versus baseline.

Predicting ICU Mortality

- Built a logistic regression model to predict 30-day ICU mortality using engineered features (instability scores, crossover counts, time-windowed statistics); applied KNN imputation and best-K feature selection, tuning regularization and decision thresholds to balance AUROC and F1-score.

AWARDS & TEACHING

- James B. Angell Scholar** — University of Michigan (March 2025); awarded to students with a perfect 4.0 term GPA
- University Honors** — University of Michigan (×4: Fall 2023, Winter 2024, Fall 2024, Winter 2025)
- Grader** — MATH 571 (Graduate Numerical Linear Algebra) and MATH 371;
- Peer Tutor**, Math Learning Center (4 semesters)